AETHERION MARK 1

Vacuum Gradient Propulsion System

Laboratory Prototype — Technical Brief

RTM-Based Propellantless Thrust Generation
Topological Capacitor Architecture
TPH/OMV Hybrid Operation

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EXECUTIVE SUMMARY

Program Objective

Demonstrate laboratory-scale vacuum gradient propulsion based on the Rhythmic Transport Model (RTM). The Mark 1 prototype validates fundamental physics before scaled development.

Key Parameters

Prototype Mass:	250 g
Core Diameter:	40 mm
Overall Height:	85 mm
Operating Temp:	293 K (ambient)
Input Power:	5-50 W
Target Thrust:	100-500 nN

Operating Principles

1. Zero-point energy storage in metamaterial
2. Asymmetric strain wave injection (TPH)
3. Ponderomotive force rectification (OMV)
4. Net unidirectional thrust generation

No propellant mass ejected.

Status: Fabrication phase. First thrust measurements scheduled Q2 2027.

PHYSICAL SPECIFICATIONS

Cross-Section (Axial)

```
        ↑ THRUST
      ┌─────────┐
     / NOZZLE    \      α=2.5
    /   CONE     \
   ┌─────────────────┐
   │  GRADIENT       │   α=2.0→0.5
   │   ZONE          │   (15 layers)
   ├─────────────────┤
   │ ACCUMULATOR     │   α=0.5
   ├─────────────────┤
   │▓▓ PIEZO ▓▓▓▓▓▓▓▓│   8× PZT-5H
   ├─────────────────┤
   │  BASE PLATE     │   α=2.5
   └─────────────────┘
```

Component Specifications

Parameter	Value	Tolerance
Overall Height	85 mm	±1 mm
Overall Diameter	60 mm	±0.5 mm
Total Mass	250 g	±10 g
Core Mass	50 g	±2 g
Core Diameter	40 mm	±0.1 mm
Number of Layers	23	—
Layer Thickness	0.3-0.5 mm	±0.05 mm
Gradient Strength	$\nabla\alpha = 200\ m^{-1}$	target

OPERATING PRINCIPLES

RTM Propulsion Mechanism

1. STORAGE

Static metamaterial gradient stores zero-point vacuum energy as structural stress. No net force in equilibrium state.

2. RELEASE (TPH)

Piezoelectric pulses create traveling strain waves. Asymmetric release generates directional impulse.

3. RECTIFICATION (OMV)

Continuous oscillation produces DC thrust via quadratic force-gradient relationship. $F \propto (\nabla\alpha)^2$

Fundamental Relationship

$$P = \gamma \cdot \nabla\alpha \cdot \nabla\phi$$

where P = momentum density, α = coherence parameter, ϕ = vacuum flux

Thrust scales with gradient steepness ($\nabla\alpha$) and metamaterial quality. No propellant mass required.

PROJECTED CAPABILITIES: PROPELLANTLESS OPERATION

Thrust Generation Without Mass Ejection

If experimental validation succeeds, the Aetherion system would enable:

Primary Characteristics:

- Continuous thrust without propellant consumption
- Unlimited operational duration (power-limited only)
- No visible exhaust plume or reaction mass
- Thrust vector independent of vehicle orientation
- Static hover capability without aerodynamic surfaces

Eliminates the Tsiolkovsky limitation.

Conventional Limitation

Chemical rockets:
~90% propellant mass

Ion drives:
Limited Δv by fuel

Aetherion:
0% propellant mass

Mark 1 Validation Target: 100-500 nN thrust at laboratory scale.

PROJECTED CAPABILITIES: ACCELERATION CHARACTERISTICS

High-G Maneuverability Potential

Theoretical Acceleration Profile:

The RTM propulsion mechanism operates via field interaction rather than momentum exchange. This implies several non-conventional acceleration characteristics:

- Instantaneous thrust vectoring (no mechanical gimbal delay)
- Rapid direction reversal without momentum penalty
- Angular maneuvers decoupled from linear trajectory
- Potential for acceleration beyond conventional g-limits*

**Inertial effects on payload/occupants remain subject to investigation.*

Note: Mark 1 measures nano-Newton thrust. Acceleration profiles are extrapolated from theory for scaled systems.

PROJECTED CAPABILITIES: OPERATIONAL SIGNATURES

Minimal Observable Emissions

Conventional Propulsion

Thermal Signature:
High IR emissions from exhaust

Acoustic Signature:
Sonic boom at supersonic speeds
Turbine/combustion noise

Visual Signature:
Exhaust plume, contrails
Plasma ionization at hypersonic

Aetherion (Projected)

Thermal Signature:
Minimal (piezo waste heat only)
No high-temperature exhaust

Acoustic Signature:
Near-silent operation
No atmospheric compression wave

Visual Signature:
No exhaust plume
No plasma formation

Implication: High-velocity operation without conventional aerodynamic/thermal signatures.

PROJECTED CAPABILITIES: MULTI-DOMAIN OPERATION

Seamless Transition Between Operating Environments

Environment-Independent Propulsion:

RTM propulsion interacts with vacuum structure rather than ambient medium. Theoretical implications:

- Identical thrust generation in atmosphere, vacuum, and submerged environments
- No performance degradation from atmospheric density variations
- Transition between air/water/space without propulsion system reconfiguration
- No cavitation effects in liquid medium (no boundary layer interaction)

ATMOSPHERIC	EXOATMOSPHERIC	SUBAQUEOUS
Full thrust authority No lift surfaces required	Vacuum operation No propellant depletion	No drag penalty No cavitation

Single propulsion architecture for all operational domains.

PROJECTED CAPABILITIES: VELOCITY CONSIDERATIONS

High-Velocity Operation Without Conventional Constraints

Conventional High-Speed Limitations (Not Applicable to RTM):

- Aerodynamic heating from atmospheric friction
  → *RTM: No boundary layer interaction*
- Sonic boom from atmospheric compression
  → *RTM: No pressure wave generation*
- Structural limits from dynamic pressure (Q)
  → *RTM: Field-mediated momentum, reduced structural stress*

Theoretical Implication:

Hypersonic atmospheric velocities without thermal signatures or acoustic emissions typically associated with such flight regimes.

SCALING PROJECTIONS

From Laboratory to Operational Systems

Configuration	Core Mass	Thrust	Power	Application
Mark 1 (Current)	50 g	100-500 nN	5-50 W	Proof of concept
Mark 2 (Projected)	500 g	10-50 μN	500 W	Satellite station-keeping
Mark 3 (Projected)	5 kg	1-10 mN	5 kW	Small craft propulsion
Mark 4 (Projected)	50 kg	0.1-1 N	50 kW	Vehicle propulsion
Mark 5+ (Theoretical)	500+ kg	10-100 N	500+ kW	Heavy lift / transport

Thrust scales with core mass and gradient quality. Power efficiency improves with scale.

All projections contingent on successful Mark 1 validation. Scaling relationships derived from RTM theory.

CURRENT STATUS & NEXT STEPS

Program Timeline

PHASE 1	PHASE 2	PHASE 3	PHASE 4
Design Complete	Fabrication	Testing	Validation

Q4 2025 Q1 2026 Q2 2026 Q3 2026

← CURRENT

Upcoming Milestones:

- ☐ Metamaterial core fabrication completion ..... March 2026
- ☐ Piezoelectric array integration ..... April 2026
- ☐ Torsion balance calibration ..... April 2026
- ☐ First thrust measurement campaign ..... May 2026
- ☐ Independent verification (if positive result) Q3 2026

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DISTRIBUTION STATEMENT

This document contains technical information relating to experimental propulsion research.

Distribution authorized to personnel with appropriate need-to-know.

All projections contingent on experimental validation.

Advanced Propulsion Research Division **LABORATORY 7Z**